مدارس الإمارات الوطنية
Emirates National Schools

Grade: 4 | Term 1, Week 12 (16 Nov to 20 Nov) | TCD: Future Skills 4: Design, Tools and Simple Solutions | Session length: 1 hr 30 min

T1 WEEK: 12	**Lesson 1: Mission Route Challenge**
Standard(s):	Technology: plan, test and debug algorithms; Math: 4.MD.C.5 (angles), grid coordinates
Literacy Strategies:	Route planning sheet with grid letters and numbers; debugging log with sentence frames: The robot went ... but I wanted ... so I changed ...
Assessment:	Route plan and debugging log; mission success; live quiz on functions, angles and loops on the TCD website.
WALT (Learning Target):	We are learning to plan, code and debug a multi-step route around obstacles.
WILF (Success Criteria):	I can plan a route on a grid before I code it. I can use loops, functions and angle blocks in one program. I can find the exact block where my code went wrong. I can record my test and my fix in a log.
Targeted Vocabulary	route, grid, obstacle, loop, function, test, bug, debug log
Resources	Matatalab Pro Sets, Map mats with grid, Obstacle pieces, Landmark picture cards (Dubai Frame, Museum of the Future, Burj Khalifa, Al Fahidi), Route planning sheets, Coloured pencils Accompanying resource: Grade4_Week12_Presentation.pptx (slides, hook video and live quiz code on the TCD website), Grade4_Week12_Worksheet and Grade4_Week12_Lesson_Plan.
National Identity Integration	Plan a route through a map of a UAE city: start at the Dubai Frame, avoid the obstacles and finish at the Museum of the Future.
AI/Digital Learning	Matatalab Pro Sets; hook video; live quiz on the TCD website.
Direct Instruction: I Do	Model route planning on the grid sheet (slide 6). Recap loops and functions and when to use each (slide 7). Model debugging with a deliberate mistake (slide 8).
Guided Practice: We Do	Class plans one short route together on the board and predicts the code. Pairs set up their mat with landmarks and obstacles (slide 9).
Independent Practice: You Do	Pairs plan, code, test and debug their mission (slide 11). Challenge levels on the board (slide 12). Live quiz on functions, angles and loops (slide 13).
Differentiated Instruction: Beginning / On / Above	**Beginning: Route with two obstacles and a planning sheet with the turns already marked.** **On: Route with four obstacles, using a loop or a function.** **Above: Route that visits two landmarks before the finish, using a loop and a function, in the fewest blocks.**
Closure/Reflection	Pairs share one bug they found and how they fixed it. Exit ticket (slide 14). Pack away.
Inclusion / EAL Support	☑ Visual Supports ☑ Chunked Instructions ☐ Reduced Language Load ☐ Model / Worked Example ☐ Pre-teach Vocabulary ☐ TA/LSA Support ☐ Other: ☐ Extra Processing Time ☐ Text Read by TA/LSA ☑ Flexible Grouping ☐ Assistive Technology ☐ Movement / Sensory Break ☑ Scaffold / Sentence Starters

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G&T / Extension	☑ Extension / Challenge Task ☐ Real-World Application ☑ Higher-Order Questioning ☐ Curriculum Compacting ☐ Depth & Complexity Task ☐ Above-Grade-Level Texts ☐ Independent Inquiry ☐ Student-Led /Passion project ☐ Leadership / Peer Coaching ☑ Open-Ended Task ☐ Other: _____